

Distributing Synthesis Across Audience Devices

The Technics, Ethics, & Aesthetics of Browser-based Co-located Network Music Performance

Thomas E Capogreco

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Abstract

For those of you who hear a voice when you read these words, and those of you who can conjure one – there is a sense in which we might regard the audio you hear to be the result of distributed synthesis.

Contents

1	Confluences of Practice & Constitutive Encounters	2
1.i	Brian Crabtree & Kelli Cain	2
1.i.a	Open Source & the Deployment of Utopia	4
1.i.b	Multiple Lineages	5
1.i.c	Mycelial Creativity	5
1.ii	Alex McLean	5
2	References	6

1 Confluences of Practice & Constitutive Encounters

This chapter aims to survey the creative practitioners with whom I have forged constitutive identifications, and to explicate the sets of commitments those identifications have implicitly subscribed me to. Although communities are a part of this landscape of practice, I have avoided framing this section as “Community of Practice” to allow for the way in which we might understand this community relation to be the implied *subject* of instrument design (and experimental music more broadly), and how for many of these practitioners, the cultivation of exactly such communities of practice can be more usefully understood to be a product of their creative practice *a posteriori*, rather than its *a priori* context.

1.i Brian Crabtree & Kelli Cain

Encountering the work of *monome* for the first time felt like a breath of fresh air. Although photos of their hardware music controllers (which began appearing on gear forums around 2008) depicted a brutally minimalist design logic, impressive craftsmanship, and well-considered ergonomics, what was most striking about these objects was how they seemed to embody a very specific design attribute – they were *open-ended*. I would like to spend a moment unpacking what this means, and explaining why I believe this approach situates monome’s design practice within what Braidotti defines as the *critical post-humanities*. (Braidotti, 2019)

Human-centered design practices inevitably begin with *empathy* – a fleshed-out picture of how and why the end-user will engage with the object of design. (Norman, 2013) The logic

being that by holding this imaginary user strongly in mind through the design process, the designer can let their empathy for that imaginary user guide the creative process in such a way that the resolved design will cater to well for other users that fit that description. In this way we can understand that designed objects always carry with them a *subject position*: the shape of the type of human the user needs to be (or, importantly, *become*) in order for the designed object to be useful (or meaningful) to them. (DiSalvo, 2012; Morton, 2011; Suchman, 2007)

The problem for music instrument design is two-fold. 1) the field of music gear is highly commodified, placing specific pressures on the subject position implicit in commercial design processes (Bates & Bennett, 2025; Théberge, 1997), and 2) this subject position becomes especially conspicuous within the parameters of a music performance ritual. (Frith, 1998; Hardjowirogo, 2016; Small, 1998) In the case of *musical expression*, it is exactly *who* the performer becomes when they interface with the musical instrument that is extruded into the semantic field as the signifier of a human. (Auslander, 2021; Souza, 2017)

What seemed to make the encounter with monome so refreshing was a kind of teleological agnosticism – the way the subject position implied by these objects seemed devoid of the regular assumptions regarding what they should be used *for*. The mechanics of the object’s affordances – both physical (button presses, encoder rotations, LED brightness levels, etc.) and connective (open sound control), were laid bare

We can understand this conceptual manouvre to be both a re-centering of systems level thinking, as proposed by Bánáthy (1996), and a simultaneous, reciprocal de-centering of the ossified, vetruvian human figure, as described by Braidotti (2019). At a systems level, we can trace what is embedded to be a technological orientation inherited from the project of *open source*. At the subject position level, I would like to argue that this teleological reduction embeds an expanded creative horizon, furnished with an extended set of possibilities regarding what *work* can be.

1.i.a Open Source & the Deployment of Utopia

... the sharing of a creation's source code is ... an anarchic act of setting free, sending off, and waiting to see how an idea or instrument may evolve in the hands and minds of an international community.

— Crabtree (2024, p. 467)

It is my contention that open source, like *synthesis*, is best understood as a material-discursive practice. Both practices implicitly seek to untether the project of modernism from the project of capitalism, inasmuch as they represent an entanglement of technical knowledges and production logics with discursively responsive sets of normative commitments – a feature notably abstracted to near-oblivion by the day-to-day mechanics of capitalist work relations. (Fisher, 2020) This normative freedom sits uncomfortably amongst the predatory injunctions of capital which inevitably function to exploit and extract value from exactly this system's capacity to facilitate autonomous, and therefore unpaid, forms of work.

In this way we might understand the material plane of this material-discursive practice to display the properties of a complex system, replete with both structural vulnerabilities, often resulting in sudden shifts of seismic proportions; and adaptability, which we must sensibly attribute to the tenacity of the utopian commitments structuring its discursive articulations. (Holland, 2003) Because of the combined effect of its unpredictability and tenacity, much discourse has emerged regarding the ways in which FOSS infrastructure articulates with an economic field dominated by massive technology corporations (Asparouhova, 2020), and how open source nomenclature can produce paradoxically extractative material effects (Tkacz, 2015).

It is with the complexity of this context in mind that we should understand monome's commitment to open source as not merely an *anarchic* gesture, as Crabtree himself describes it (2024, p. 467), but also as a de-centering of the locus of design from a human orientation, to a systems orientation. (Suchman, 2007) Rather than insinuating that the object can take the user all the way to some enclosed, pre-fabricated utopia, the object, in monome's own words, "does nothing". Any utopian potential the user might see in the object then becomes

contingent on the user inhabiting an empowered subject position either equipped with forms of technological literacies required to unlock that potential, or, importantly, the disposition to learn them.

In this way we can therefore understand utopia to be deployed not as marketing, but in the set of commitments that constitute the objects’ subject position. In addition to implicating specific forms of work, these utopian logics, now housed in the constitutive commitments of users’ subjectivities, have the opportunity to become operable in a sustained manner, at the point of these material objects’ discursive articulation, via the action of their users’ language – a feedback loop which, as mentioned earlier, is important in imbuing the material-discursive system with the attribute of adaptability.

1.i.b Multiple Lineages

*... there are multiple lineages
and paths being traveled herein.*

— Crabtree (2024, p. 465)

We might trace monome’s commitment to open source back to when the pair met at University of California San Diego. Crabtree studied under the tutelage of Miller Puckette and Tom Erbe, who are notable here not merely because of the way in which their design thinking influenced monome, but because of their direct influence on my own creative practice and the project of distributed synthesis more specifically.

1.i.c Mycelial Creativity

1.ii Alex McLean

2 References

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